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Towards intelligent food safety: Machine learning approaches for aflatoxin detection and risk prediction

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ABSTRACT

Aflatoxins pose a grave threat with potentially devastating health effects that go unnoticed in our everyday food supply, especially foods such as peanuts, maize, and spices, particularly in the tropics and sub-tropics, where climatic conditions are conducive to their production. Existing methodologies for aflatoxin determination remain costly and time-consuming, preventing their implementation on a practical level for fast and widespread deployment through real-time monitoring. This review provides a landmark, integrative perspective of how artificial intelligence (AI) in its numerous forms has advanced aflatoxin detection and quantification across agricultural systems. In particular, this review considers the integration of food science and AI by understanding the application of supervised, unsupervised, and reinforcement learning to spectral, image, and behavioral data analysis used to inform aflatoxin detection and quantification. Combined with state-of-the-art applications such as AI smartphone-based diagnostics, clever storage systems, and deep learning models for image analysis, this review examines various cases and evaluations of developed models, addressing critical real-world challenges such as sparse data, generalization across food matrices, and regulatory transparency. Ultimately, the review addresses the willingness to adopt evolving AI strategies and looks to the future for faster, wiser, and more accessible aflatoxin detection methods for more significant public health protection and sustainable food systems. Finally, regardless of whether you are a uniquely positioned researcher investigating the development of new models, a policymaker developing food safety regulations, an academic designing curriculum, or a scientist inquisitively exploring the next generation of food technologies, this article is a timely and convenient place to access knowledge leading toward safer, AI-powered food systems.

1. Introduction

In a time where the world is coping with an ever-changing set of public health issues, ranging from viral pandemics and their sequelae to antibiotic resistance, there remains an overlooked and perniciously invisible threat that has long been overlooked: aflatoxin contamination in our food supply (Magdalena Pisoschi et al., 2023)(L. Wang et al., 2022). While it is understandable for the focus to shift from one public health threat to another, many do not realize that something as simple as a bowl of cornflakes or a handful of peanuts could harbour an active disease-causing toxin known as aflatoxin. (Dadmehr & Korouzhdehi, 2023; Pérez-Fernández & de la Escosura-Muñiz, 2022). Aflatoxins are considered some of the most toxic, naturally occurring compounds known to mankind. Yet, these poisons endure and quietly circulate through our international food systems, particularly in vulnerable populations with climate and infrastructural burdens. (Miklós et al.,

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2020; Yan et al., 2020). Aflatoxins are toxic secondary metabolites produced by two species of fungus, Aspergillus flavus and Aspergillus parasiticus (Alameri et al., 2023). The molds that produce aflatoxins thrive in warm, humid conditions that are particularly relevant in the tropics and subtropics. Still, with the escalation and impact of climate change, they are proliferating in areas that have never had to contend with these fungal infections. There are several forms of aflatoxins, B1, B2, G1, G2, and the M-series (M1 and M2). However, Aflatoxin B1 is the primary toxin of concern, possessing proven carcinogenic toxicity (Yadav et al., 2021)(Thurner & Alatraktchi, 2023). Chronic exposure to Aflatoxin B1 in small amounts can have detrimental effects, leading to liver cancer, immune suppression, and growth stunting among children. Acute exposure is infrequent. However, aflatoxicosis can be lethal. Perhaps the most significant detriment to its toxicity is that aflatoxins are invisible; they do not change the taste, smell, or appearance of food, so consumers cannot identify them. (Zhang & Guo, 2022).

The global implications of aflatoxin contamination extend beyond health alone. The FAO has reported that approximately 25 % of food crops grown globally are impacted by mycotoxins, with aflatoxins representing the most toxic group (Shabeer et al., 2022). Thus, aflatoxin contamination has several implications, including, but not limited to, public health burdens and resulting economic losses (Babu et al., 1994), especially in agricultural-based economies (Abdeta & Milki, 2023; Wu, 2015). Consequences may include losses due to export rejections, consumer distrust, or even increased financial burdens to healthcare systems. For that reason, governing bodies, such as the World Health Organization (WHO), the U.S. Food and Drug Administration (FDA), the European Food Safety Authority (EFSA) and the Food and Agriculture Organization (FAO), have established stringent maximum limits for aflatoxins in food and feed products (Fuffa & Urga, 2001). Enforcing these limits presents significant challenges to low-resource settings. Historically, scientists and food safety organizations have developed several techniques for aflatoxin detection. Conventional techniques (i.e., Thin Layer Chromatography (TLC), High-Performance Liquid Chromatography (HPLC), and Gas Chromatography-Mass Spectrometry (GC-MS)) provide high levels of accuracy and sensitivity. Immunoassay-based detection techniques (e.g., Enzyme-Linked Immunosorbent Assays (ELISA), Lateral Flow Immunoassays (LFIA), etc.) have also been utilized for general fast screening (Frizzarin & Duarte, 2012; Sánchez-Bayo et al., 2017; Sani et al., 2014; Sudini et al., 2012). While various techniques can be effective under pre-controlled experimental conditions, they are also costly, time-consuming, and often require personnel with training, education, and sophisticated equipment. Furthermore, laboratory detection techniques are typically based upon central laboratory models, which often limit the feasibility of real-time or immediate detection of aflatoxins, especially in rural or developing settings that often are the highest risk for aflatoxins (Lifan, 2014).

The disparity between the available detection methodologies and the need for expedited, inexpensive, and scalable alternatives has generated opportunities for innovative research (Miller, 2023)(Kharbas et al., 2024, pp. 1–6). Recently, learning ML and AI methods has gained momentum in this area. The ability of AI and ML algorithms to dissect large and complex datasets, detect meaningful patterns, and make predictions is revolutionizing how we will approach food safety (Liu et al., 2023)(X. Wang, Bouzembrak, Lansink, et al., 2022). Machine learning algorithms can use spectral, chemical, and image data to detect aflatoxins and quantify them with unprecedented precision. In addition, AI methods can automate decision-making based on prior observations, learn from new data, and provide real-time monitoring with newly developed Internet of Things (IoT) sensors. For example, a storage facility with smart sensors that detect early signs of fungi, perform on-the-spot analysis with local AI models, and alert farmers or food safety officers before fungi infect crops is now conceivable (Haghbin et al., 2023; Przybył et al., 2020). In fact, these systems are being pilot-tested in various parts of the world now. Applications for AI may take the form of regression models that predict the concentration of toxins, classification algorithms that identify contaminated vs. accurate samples, and deep learning models analyzing images of crops for infected areas. The applications of AI in aflatoxin detection are developing, and the potential is numerous. Current research directions include reinforcement learning, blockchain, and innovative packaging systems to increase consumer transparency, traceability, and trust in food systems.

The implications of this technological shift extend far beyond laboratories and academia. It can improve lives. Access to AI-powered testing is necessary for smallholder farmers to prevent catastrophic financial losses from spoiled and/or contaminated crops (Verma et al., 2022). For exporters, it improves the reliability of meeting international safety standards. It adds a layer of assurance about safe food and less risk of chronic disease for consumers, particularly food-insecure populations. More broadly, AI provides governments and policymakers with practical data for predictive analytics, which aids in resource allocation, pinpointed interventions, and proactive risk mitigation in the fight against aflatoxin. What marks this review as different from others is its integrative nature. Many reviews focus exclusively on conventional detection methods; others focus on one or two components or aspects of AI in agriculture. No one has compiled a complete picture of machine learning being applied to the problem of aflatoxin detection (Adamashvili et al., 2024). This review builds on case studies, emerging technologies, and real-world applications and considers the challenges, data limitations, and expectations for explainable AI from a regulatory perspective. This review intends to provide a knowledge repository for researchers, engineers, policymakers, and industry professionals by bridging food and computer sciences. As we go deeper into an age driven by digital transformation, the convergence of AI and food safety is timely and greatly needed. The threat of aflatoxins is ancient, but the tools to combat them contain innovations that are anything but. In the following sections, we will describe how machine learning is changing aflatoxin detection and, ultimately, share how we may advance food system safety, intelligence, and sustainability by mobilizing innovation.

2. Machine learning methods for aflatoxin detection

ML is a class of AI that allows systems to interpret data sets, identify patterns, and make informed judgments without being programmed directly. Unlike computational models, which previously required a set of fixed instructions and demonstrated limited capabilities in understanding complex environments, an ML system learns continuously from many datasets, increasing its predictions over time. ML systems can be employed to address complex multicultural public health situations like food safety, which may involve looking beyond a specific location or list of parameters in understanding contamination (van der Schaar et al., 2021). In this review, attributing ML systems to aflatoxin contamination detection represents an exciting new direction with some serious considerations due to the challenges of detecting aflatoxins in food, which places demands on the technology (the identification, degree of severity, and contamination and assessment of risk) and the need for quicker and cost-effective detection. The use of ML systems for detecting aflatoxins in food offers significant advantages compared to traditional detection methods, which often include chromatography and immunoassays that typically require expensive laboratory settings, experienced personnel, and time processes. The nature of developing reagents, especially for immunoassays, leads toward larger-scale and real-time monitoring solutions that contribute to existing knowledge on food safety practices (traceability, transportation from the source to retailer, and ecology conservation). Enterprise software solutions increase real-time detection, measuring total aflatoxin, aflatoxin B1, and aflatoxin B2, and reflect user experience, indicating contamination levels or degree of risk can be identified in real-time environments without prescribed laboratory facilities common in aflatoxin testing (Han & Gao, 2019). Integrating ML into the detection strategy offers significant opportunities for efficient testing, interfacing data environments, and reducing human error in situ food safety monitoring. This integration can improve food safety in real-time and confidence from the consumer's perspective, making their recommendations based on lab-induced data to larger commercial aspects of agriculture. To understand the relationship of ML in aflatoxin detection, the first important elements of ML can be understood in this context, and each of the components contributes to recognizing how they are related to the benefits of ML.

2.1. Types of machine learning and their relevance to aflatoxin detection

ML can be divided into three major categories: Supervised, Unsupervised, and Reinforcement Learning (shown in Fig. 1). Each of these types of learning can be implemented for distinct applications, with advantages and disadvantages to each kind, and their applicability in the detection of aflatoxin will differ depending on the problem to address. Supervised learning is one of the most commonly applied ML methods by training on labeled data; training with labeled data provides known output labels with each input data point, helping the model learn by example (T. Jiang et al., 2020). Therefore, it is evident that supervised learning can be beneficial for aflatoxin detection, as models can be trained on datasets with known levels of contamination with aflatoxin. Within supervised learning, classification models are used to determine whether a sample is contaminated or not, while regression models predict aflatoxin concentration based on features such as spectroscopic or chromatographic data. Standard algorithms for the classification of food samples based on contamination status include Support Vector Machines (SVMs), Random Forests (RFs), and Convolutional Neural Networks (CNNs). Standard regression methods to quantify levels of aflatoxins include Linear Regression, Support Vector Regression (SVR), and Random Forest Regression (RFR) (Zhao et al., 2015).

Conversely, unsupervised learning does not rely on any form of labeled data and is mainly employed to identify latent structures or patterns within datasets. This is especially advantageous when trends regarding contamination or contamination outbreaks have not been well-reported or even when very large datasets belong to the agricultural area. For aflatoxin detection, unsupervised learning is applied through clustering and dimensionality reduction. Clustering methods like K-Means and Hierarchical Clustering can categorize food samples by detecting similarities and assist researchers in grouping contaminated and non-contaminated products without having had a labeled, categorized dataset to begin with (Chinchuluun et al., 2009). Dimensionality reduction techniques, such as Principal Component Analysis (PCA) and Autoencoders, are crucial in processing large and complex datasets, especially those derived from spectroscopy and biosensor readings. These methods enhance model performance and speed up processing time by reducing the number of input variables while retaining meaningful information. Reinforcement learning (RL), the third main form of ML, is a learning process through dynamic interactions. In RL, an AI agent interacts with an environment and learns optimal actions by receiving rewards or penalties for correct and incorrect decisions (Shakya et al., 2023). While RL is often associated with robotics, gaming, and autonomous systems, it shows potential in food safety and aflatoxin detection. For example, RL can be helpful for real-time monitoring of aflatoxin contamination. Self-learning models that employ RL principles will autonomously improve confidence in their predictions by learning new contamination patterns through a reward mechanism. RL models are especially valuable in the context of smart IoT-enabled food storage systems. In this theory, RL-based AI systems will continuously predict the possibility of contaminated food and adjust storage conditions to the conditions necessary to prevent fungal growth and aflatoxin contamination. RL also has potential utility in the optimization of screw supply chains intended to continually provide consumers with aflatoxin-free products while limiting economic impact and health risk.

Of the three types of ML, supervised learning is by far the most common for research and applications of aflatoxin detection. Models utilizing supervised learning can leverage labeled data for training stages and are helpful for classification and regression tasks, which are imperative in predicting levels of aflatoxin contamination. Researchers employ Standard supervised learning models for classifying contaminated food items: Support Vector Machines, Random Forests, and Deep Learning models. Regression-based approaches help provide precise quantification for levels of aflatoxin concentration (Castano-Duque et al., 2022). Although unsupervised learning is typically not utilized as a primary means to detect aflatoxins, the technique will have an important secondary function to assist with preprocessing datasets and exploratory analysis. Unsupervised learning can discover contamination patterns without needing labeled datasets, improving the refinement of the input data and facilitating better classification model performance. As ML methodology develops, the prototypes and implementations of supervised machine learning algorithms will increasingly combine with unsupervised learning to produce hybrid models for improved detection. Lastly, although still developing, reinforcement learning has the potential for real-time, decision-making-based intelligent monitoring of aflatoxins in food safety. As artificial intelligence and intelligence-driven food monitoring technology advances, reinforcement learning will support and share datasets to improve risk assessment, on-site detection of contamination, and develop self-learning

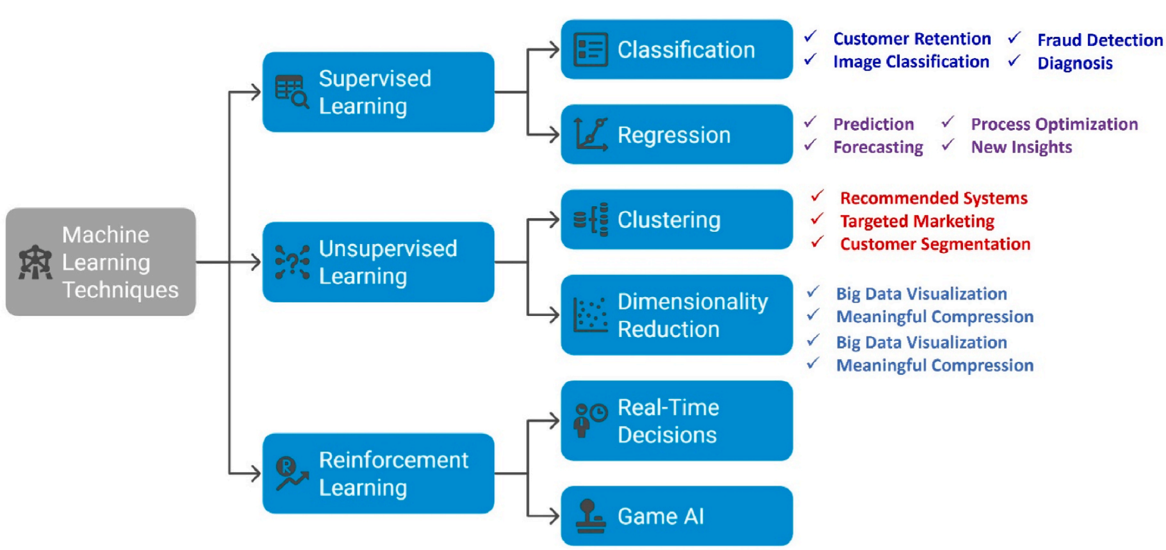

Fig. 1. Overview of machine learning techniques and their applications.

mitigation strategies for aflatoxins (Gavrilov & Lee, 2007).

2.2. Understanding how machine learning algorithms work for aflatoxin detection

Having discussed the different types of ML being used to detect aflatoxin, let's take a moment to take a closer look at how these model's function. ML algorithms utilize a learning process analogous to humans learning from experience, recognizing patterns, making predictions, and refining knowledge. Whereas humans often rely on their intuition to make intelligent decisions, ML approaches employ mathematical models that can process large datasets and make intelligent decisions. To begin to understand how ML can be deployed for aflatoxin detection, we will consider three categories of techniques: regression algorithms (which

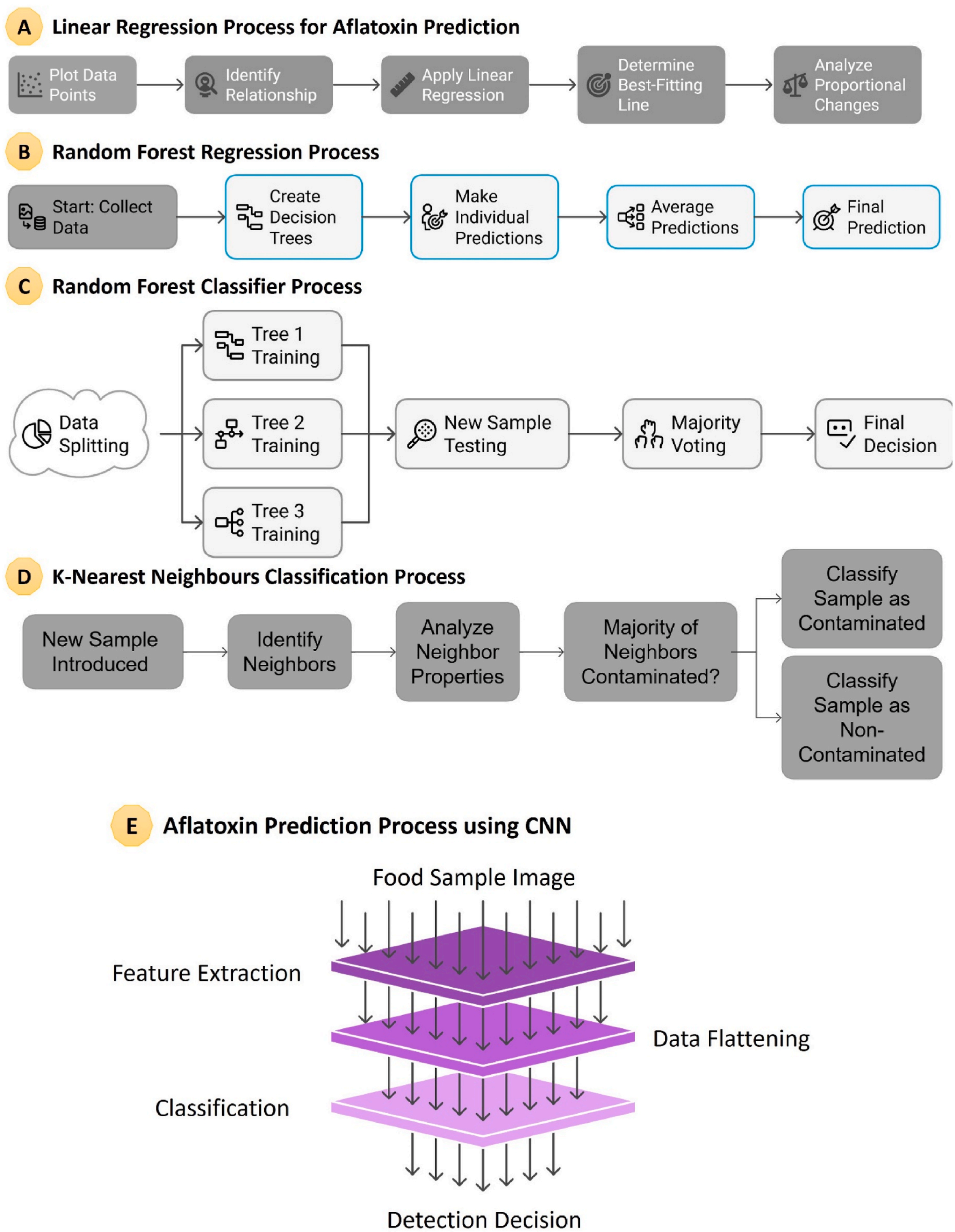

Fig. 2. A) Linear regression process for aflatoxin prediction. B) Random forest regression process for aflatoxin prediction. C) Random forest classifier process for aflatoxin prediction. D) K-Nearest Neighbors classification process for aflatoxin prediction. E) Aflatoxin prediction process using CNN.

predict aflatoxin levels in food samples), classification algorithms (which assess whether the sample is contaminated or not), and deep learning models (which assess images for contamination). Each technique has its unique functionality for constructing an intelligent automated detection.

2.2.1. Regression algorithms: predicting Aflatoxin levels

Imagine you're planning a picnic and want to predict the temperature for the day. You check past temperature trends, humidity, and wind patterns to estimate the weather conditions. This is precisely how regression algorithms work in machine learning; they take input data and predict a numerical outcome (M. L. Bhaiyya et al., 2023). In aflatoxin detection, regression models predict aflatoxin concentration in a food sample based on measurable features like spectroscopic data, chemical composition, or sensor readings. Let's break down a few key regression techniques in this domain: 1) Linear Regression: Linear regression is like drawing a straight line through a set of points on a graph. If you were to plot aflatoxin concentration on the Y-axis and sensor readings on the X-axis, a linear regression model would find the best-fitting line that represents the trend. This model assumes that as one factor changes, the aflatoxin level changes proportionally (as shown in Fig. 2 (A)) (Maulud & Abdulazeez, 2020). Though simple, linear regression is practical for straightforward relationships where aflatoxin concentration increases or decreases predictably with changes in specific properties. 2) Support Vector Regression (SVR): While linear regression works well for simple cases, real-world aflatoxin contamination is rarely that straightforward. SVR is a more advanced regression technique that fits a line and creates a "margin of tolerance," allowing it to handle more complex, non-linear relationships. This makes SVR ideal for predicting aflatoxin levels when food samples show irregular contamination patterns (Parbat & Chakraborty, 2020). 3) Random Forest Regression: Imagine you're asking multiple experts for their opinion on a food sample's contamination level. Instead of relying on one person's answer, combine everyone's insights to get the best possible prediction. Random Forest Regression (RFR) (as shown in Fig. 2 (B)) works similarly, it creates multiple decision trees, each making its prediction, and then averages them to arrive at the most reliable result (Borup et al., 2023). This method is widely used in aflatoxin detection because it efficiently handles complex, high-dimensional data. Using these regression techniques, machine learning can accurately predict aflatoxin concentration in food samples, helping food safety authorities take necessary actions before contamination reaches dangerous levels.

2.2.2. Classification algorithms: sorting safe vs. contaminated food

Consider the case of an airport security scanner, which uses a model to classify baggage as safe or suspicious. The scanner uses several features, including the bags' size, shape, and density, to analyze the luggage and decide whether to seek additional inspection (Garg et al., 2022). Similarly, ML classification algorithms examine the features of sample data (in our case, food samples) and classify them as contaminated or non-contaminated. Some standard classification techniques applied in aflatoxin detection include: 1) Support Vector Machines (SVM): Imagine trying to draw a line between two collections of dots, one with safe food and the other with contaminated food. The purpose of an SVM is to find the "best" dividing line (also referred to as a boundary) that maximizes the distance between the two groups (Bennett-Lenane et al., 2022). SVM is beneficial for distinguishing between aflatoxin-free and contaminated food samples because even though the samples may have subtle differences, the SVM can classify the samples and separate the food products. 2) Random Forest Classifier: Similar to Random Forest Regression, the Random Forest Classifier grows multiple decision trees, each using an independent random data selection as input for training. When a new food sample is tested for food safety, each tree votes on whether the sample is contaminated, and the final decision is based on most of the votes (as seen in Fig. 2 (C)). The Random Forest Classifier is accurate and minimizes error, so it is often a serious candidate as a supervised machine learning model for aflatoxin detection. 3) K-Nearest Neighbors (KNN): Finding Similar Examples. When determining if a food sample is contaminated, you would want to compare it to collected samples prior instead of having to examine the sample individually. If most of the samples collected and assessed nearby are contaminated, it is reasonable to assume that this sample may also be contaminated. KNN works similarly, accurately classifying new food samples based on similarity to previously sampled food samples with 1) Likert scales (previously labeled) samples. When new data is recorded, the KNN model classifies the data based on the similarity to previously recorded food samples (see Fig. 2 (D)). It's a basic model, but it is relatively effective for small datasets. Classification models (which we will return to later) are critical for real-time food safety applications. They facilitate inspectors to make an informed decision on whether or not a batch of food should be accepted, further tested, or rejected (Kok et al., 2021).

2.2.3. Deep learning for image-based Aflatoxin detection

Next, let's advance the topic even further. What if a machine could take an image of a food sample and detect contamination as well as, or even better than, a human inspector? Enter deep learning. Deep learning is a type of ML that mimics how the human brain can solve/understand problems. In this scenario, deep learning provides a strong capability for image-based aflatoxin detection (M. Bhaiyya et al., 2024). Deep learning models can be trained on thousands of images of food samples containing aflatoxin, for instance, healthy grains vs. grains infected by aflatoxin, and use color, texture, and fluorescence patterns to discern whether contamination is likely present or not. Common deep learning architectures for aflatoxin detection include: 1) Convolutional Neural Networks (CNNs): CNNs are like the eyes of AI. Given that our brains process various parts of an image, edges, shapes, and colors, CNNs evaluate images in layers. The CNN first detects basic features (e.g., edges, textures) and later complex features (e.g, mold patterns, fluorescence patterns suggesting aflatoxin contamination). CNNs produce images interpreting the presence of aflatoxins successfully using TLC plate imaging, fluorescence microscopy, and hyperspectral imaging, as is evident in their prevalence (Fig. 2 (E)). 2) Recurrent Neural Networks (RNNs): CNNs are great for taking images of static images; when there is time-series data, RNNs are helpful. For example, assessing aflatoxin contamination over time in a storage facility is far more manageable with RNNs. RNNs can remember data points from the past and contextualize to guess risk into the future. RNNs help the producer and food safety officer make the call much earlier than when it is presented (Kılıç & İnner, 2022). With deep learning networks, the problem of detection of aflatoxins can be automated, and then subsequently, it will make it easier to scale, even more, to be useable (e.g., quality control of AI with food industries) and create economies of scale.

2.3. Performance evaluation metrics for regression and classification ML models

The evaluation of a machine learning model can be considered similar to the grading of an exam for a student. As we grade a student's performance in an exam using non-differentiating marks or a percentage to summarize how well the student performed, we evaluate the performance of a machine learning model by determining how well the machine learning model is predicting the output it was asked to predict. The evaluation metric we select depends on the machine learning task - regression or classification (Kahar et al., 2023). With regression problems where the model predicts a continuous value (for example, aflatoxin concentration in a food sample), we can use evaluation metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Square Error (RMSE), R^2 Score, and Adjusted R^2 Score. With classification problems where the model predicts a categorical outcome (for example: "contaminated" versus "non-contaminated"), we can select performance metrics such as Accuracy, Precision, Recall, and F1-Score, which attempt to measure how accurately the model is distinguishing between categories (Shirmohammadi & Al Osman, 2021).

2.3.1. Performance metrics for regression models

1) Mean Absolute Error (MAE) is one of the simplest ways to measure a model's accuracy in regression problems. It tells us, on average, how much the model's predictions differ from the actual values. The smaller the MAE, the better the model's performance. Imagine you are a food scientist trying to predict the aflatoxin concentration in a batch of maize. If the true aflatoxin level is 10 parts per billion (ppb) and your model predicts 8 ppb, the error is 2 ppb. MAE calculates the average of such absolute errors across all predictions, showing how far off the model is from reality.

2) Mean Squared Error (MSE) is like MAE, but with one key difference: it squares the errors before averaging them. This means bigger mistakes are penalized more heavily than smaller ones. For instance, if your model underestimates the aflatoxin level by 5 ppb in one sample and 1 ppb in another, squaring these values results in 25 and 1, respectively. MSE ensures that large errors significantly impact the overall score, making it useful when it's crucial to avoid significant prediction errors. 3) Root Mean Squared Error (RMSE) is simply the square root of MSE, making it easier to interpret because it brings the error back to the original unit of measurement. Think of RMSE as a refined version of MSE; it still penalizes more significant errors but provides results that are easier to compare with actual aflatoxin concentration levels. If an RMSE score is 2.5 ppb, it means that, on average, the model's predictions are off by around 2.5 ppb, making it a practical metric for evaluating real-world contamination levels. 4) R^2 Score (or the Coefficient of Determination) tells us how well a model explains the variability in the data. It is measured on a scale from 0 to 1, where 1 means a perfect prediction and 0 means the model is no better than random guessing. Imagine you are a teacher grading students' test scores. If a student gets a perfect 100 %, it means they have mastered the subject ($R^2 = 1$). If their performance is random and unpredictable, their score is closer to 0. Similarly, a high R^2 value in regression models means the model effectively captures trends in aflatoxin levels. 5) The problem with R^2 is that it always increases when more input features (variables) are added, even if those features don't improve the model. Adjusted R^2 fixes this by penalizing unnecessary complexity and ensuring that only handy features contribute to the model's accuracy. Hiring employees and adding more workers might seem beneficial, but they're just increasing costs if they aren't helping. Adjusted R^2 ensures that only meaningful variables are included in the model (Naser & Alavi, 2023).

2.3.2. Performance metrics for classification models

1) Accuracy is the simplest way to evaluate a classification model; it tells us how often its predictions are correct. For example, if a machine learning model correctly predicts 95 out of 100 food samples, its accuracy is 95 %. While accuracy is practical, it can be misleading if the dataset is imbalanced (for example, if 90 % of food samples are safe, a model that always predicts "safe" would have 90 % accuracy but fails to detect contamination properly). 2) Precision measures how many of the samples predicted as contaminated were contaminated. Imagine a food safety inspector tests 20 food samples and identifies 10 as contaminated, but only 8 are contaminated (Valero-Carreras et al., 2023). In this case, precision is 80 %. High precision means fewer false positives (wrongly marking safe food as contaminated), which is useful when discarding food unnecessarily and could lead to economic losses. 3) Recall (or Sensitivity) tells us how many truly contaminated samples were correctly identified. If 30 food samples are contaminated, but the model only detects 20, the recall is 66.7 %. High recall is essential when missing a contaminated sample, which could pose serious health risks. It ensures that as many dangerous food items as possible are detected. 4) The F1-score is the balance between Precision and Recall. The F1 score is useful when both false positives and false negatives matter. For example, if precision is high but recall is low (meaning the model catches only a few contamination cases), the F1-score helps find a good middle ground. The pictorial representation of performance evaluation metrics for regression and classification models with related formulas is shown in Fig. 3 (Naser & Alavi, 2023).

3. ML applications in aflatoxin detection and risk prediction: case studies

Over the past few years, numerous studies have demonstrated the immense potential of ML and DL in improving the detection, classification, and quantification of aflatoxin contamination in various agricultural products. The integration of spectral technologies, hyperspectral imaging, and advanced ML algorithms has not only enhanced the accuracy of predictions but has also addressed key challenges in traditional detection methods like time, cost, and the need for specialized laboratory setups.

Recent advances in ML have significantly transformed aflatoxin detection across various stages of the food supply chain. For instance, Han and Gao developed a pixel-level aflatoxin detection model using hyperspectral imaging and convolutional neural networks (CNNs), achieving over 95 % accuracy at the pixel level and over 90 % at the kernel level in peanuts, shown in Fig. 4 (A). This high-resolution detection is ideal for integration into automated sorting systems (Han & Gao, 2019). Building on the theme of spectral imaging, Zhu et al. introduced MSGhostDNN. This Siamese contrastive learning framework achieved 97.87 % accuracy in detecting aflatoxin B1 in peanuts and corn by focusing on informative wavelengths (shown in Fig. 4 (B)). It sets a benchmark for precision, adaptability, and efficient online monitoring of aflatoxins in global food systems (Zhu et al., 2024). Similarly, Deng et al. designed a dual-task deep learning model integrating microwave detection with CNNs to simultaneously assess mildew and aflatoxin B1 in wheat, reaching 100 % classification accuracy and an R^2 of 0.98 (shown in Fig. 4 (C)). This method reduces the need for separate sensors or chemical assays and streamlines the testing process (Deng et al., 2023). Extending the scope to animal health, Mei et al. used wearable triaxial accelerometers on broiler chickens to monitor behavioral changes caused by aflatoxin exposure. Models including Gradient Boosted Decision Trees (GBDT) and Random Forests identified aflatoxin exposure with over 86 % accuracy, with GBDT achieving 97 % accuracy between healthy and severely affected birds. This research represents a cross-domain application of AI in animal health and food safety (shown in Fig. 4 (D)). It highlights how wearable sensors and behavioral analytics can provide early, automated disease detection, minimizing economic loss and enhancing livestock welfare (Mei et al., 2023).

Moving toward liquid and image-based matrices, the Deng et al. study explores the potential of Raman spectroscopy combined with deep learning to detect aflatoxin B1 (AFB1) in edible oils. Two deep learning models, a CNN and an RNN, were developed for both qualitative and quantitative analysis of aflatoxin contamination. Both models achieved 100 % classification accuracy, while the RNN outperformed CNN in predicting AFB1 levels, reaching an R^2 of 0.95 and an RPD of 4.86 (shown in Fig. 5 (A)). This approach offers a rapid, non-destructive, cost-effective alternative to chromatography-based laboratory tests. It underlines how AI-powered Raman spectroscopy can transform mycotoxin detection into a portable, scalable solution (Deng, Zhang, et al., 2022). Likewise, Jiang et al. employed Gramian Angular Summation Field (GASF) encoding to convert NIR spectra into image data for CNN processing, achieving an R^2 of 0.99 and RPD of 8.3 in detecting aflatoxin B1 in moldy peanuts. This study showcases the fusion of advanced spectral imaging and deep learning for high-throughput, non-invasive aflatoxin quantification, as shown in Fig. 5 (B)). It exemplifies the shift toward intelligent spectrometry and image-based spectral encoding in agri-food tech.

(H. Jiang et al., 2023). In the context of dried fruits, Kılıç et al. used deep learning models such as MobileNetV2 and ResNet101V2 to classify aflatoxin contamination in dried figs with up to 96 % accuracy,

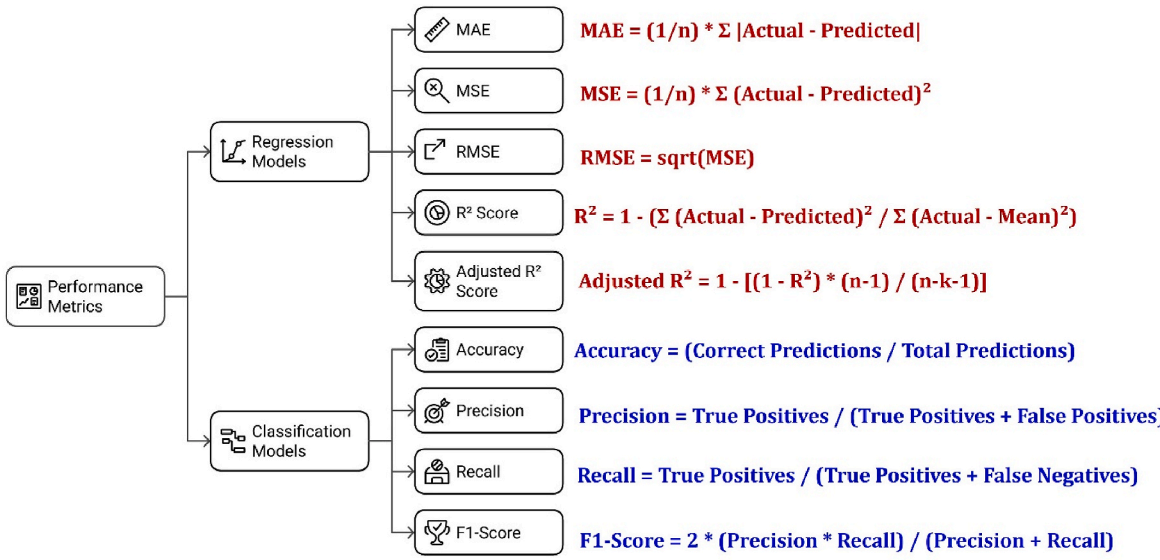

Fig. 3. Pictorial representation of performance evaluation metrics for regression and classification models.

replacing manual fluorescence methods and enhancing safety in sorting environments (Kılıç et al., 2024). Similarly, Sadimantara et al. paired fluorescence imaging with CNNs to detect contamination levels in cocoa beans, achieving up to 96.4 % accuracy, which makes the method a viable low-cost alternative to LC-MS in post-harvest stages, especially in developing regions (as shown in Fig. 5 (C)) (Sadimantara et al., 2024). To further reduce chemical dependence in processing, Evrendilek et al. combined pulsed electric field (PEF) treatment with machine learning models like Random Forests and Gradient Boosting Trees to achieve up to 99.8 % aflatoxin reduction in red pepper flakes, demonstrating the synergy between non-thermal treatment and predictive modeling (Akdemir Evrendilek et al., 2022).

Integrating mobile and portable diagnostics, Sun et al. developed a smartphone-based aflatoxin B1 detection system using YOLOv5 to analyze immunochromatographic test strips, achieving 98 % accuracy and a detection limit as low as 2.2 pg/mL. This system is portable, wireless, and adaptable across smartphone platforms, offering a fast, user-friendly point-of-care testing method. This paper showcases how AI, particularly YOLO-based models (as shown in Fig. 5 (D)), can democratize toxin detection. It signals a shift toward accessible, community-level biosensing and empowers field-level food safety without lab infrastructure (Sun et al., 2025). In the realm of ultra-sensitive biosensing, Hu and colleagues introduced a THz metamaterial sensor capable of detecting aflatoxin B2 at nanogram-per-liter levels (LOD $= 7.28 \times 10^{-11}$ mg/mL), showcasing the fusion of physics and AI for next-generation food safety tools (Hu, Zhan, Wang, et al., 2023). Complementing this, Zhu et al. proposed an unsupervised pixel-level detection framework combining convolutional autoencoders with Fuzzy C-Means clustering, followed by transfer learning-based classification models, achieving over 97 % accuracy even with limited labeled data (Zhu, Zhao, et al., 2023). Finally, Sabo et al. explored the use of biogenic volatile organic compounds (bVOCs) emitted by infected peanuts, applying machine learning on GC-MS data to detect contamination with high kernel-level accuracy (F1 score $= 0.80$). This unique biological sensing approach opens new directions for non-invasive, real-time aflatoxin diagnostics in pre- and post-harvest systems (Sabo et al., 2024). To better understand the diversity of ML approaches applied to aflatoxin detection, a comparative overview of various case studies is presented in the following Table 1. This summary highlights the spectrum of detection techniques, ML models employed, performance metrics such as accuracy or R^2, and the broader trends they represent, ranging from portable diagnostics and spectral data modeling to behavioral monitoring and smartphone-enabled sensing. It also includes the targeted agricultural product types, offering a consolidated perspective on how AI-driven technologies reshape food safety monitoring across different domains.

4. Challenges and future scope in AI-based aflatoxin detection

The rapid adoption and incorporation of AI-ML into food safety systems, particularly in aflatoxin detection, will need to reflect on the future limitations and potential transformative developments. AI-ML algorithms have successfully detected aflatoxins within various food matrices, but still need to reach maturity in operational settings. This section addresses some of the main challenges that need to be addressed. It introduces some emerging trends paving the way for future food safety solutions using artificial intelligence.

4.1. Data limitations and the need for larger training datasets

Current AI approaches for identifying aflatoxins depend largely on high-quality labeled datasets. These datasets may be available sometimes but acquiring large amounts of these labeled datasets for multiple crop types, world regions, and contamination levels is expensive and laborious. The small-sized datasets also have overfitting issues, leading to model failure if it is not likely to generalize beyond the localized real-world context. One possible solution is the development of publicly available collaborative databases to encompass aflatoxin detection data in a global comparison that addresses coverage and variability. Employing data augmentation, such as creating realistic training data using generative adversarial networks (GANs), can be employed, particularly for rare cases of contamination. Transfer learning and federated learning models make use of multiple decentralized datasets to improve model performance without exposing their confidentiality, to improve analytical power in situations where localized data is limited (Rahman et al., 2024).

4.2. Model generalization across diverse food matrices

AI models trained on certain food commodities, such as peanuts or maize, may not perform well on other matrices, such as spices, nuts, or oils. The basis for their variability in performance can be attributed to differences in surface characteristics, chemical makeup, and various imaging conditions. These models' poor generalizability to other food

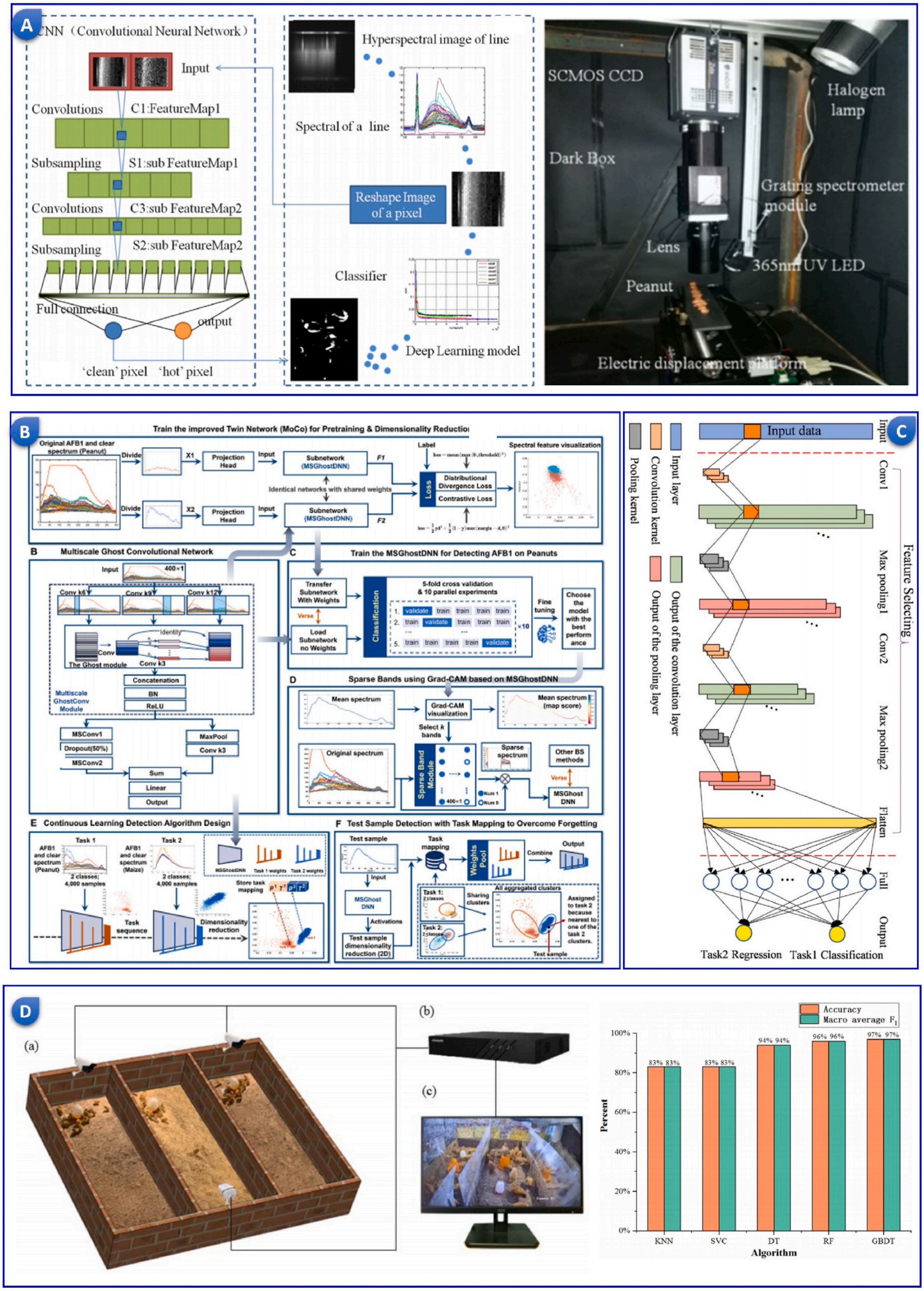

Fig. 4. A) Pixel-level aflatoxin detection based on deep learning and hyperspectral imaging, taken from (Han & Gao, 2019) with the permission of Elsevier. B) Spectral intelligent detection for aflatoxin B1 via contrastive learning based on a Siamese network, recreated from (Zhu et al., 2024) with the permission of Elsevier. C) Multi-task deep learning strategy to detect aflatoxin from wheat (Deng et al., 2023) with the permission of Elsevier. D) Identification of aflatoxin-poisoned broilers based on accelerometer and ML, taken from (Mei et al., 2023) with the permission of Elsevier.

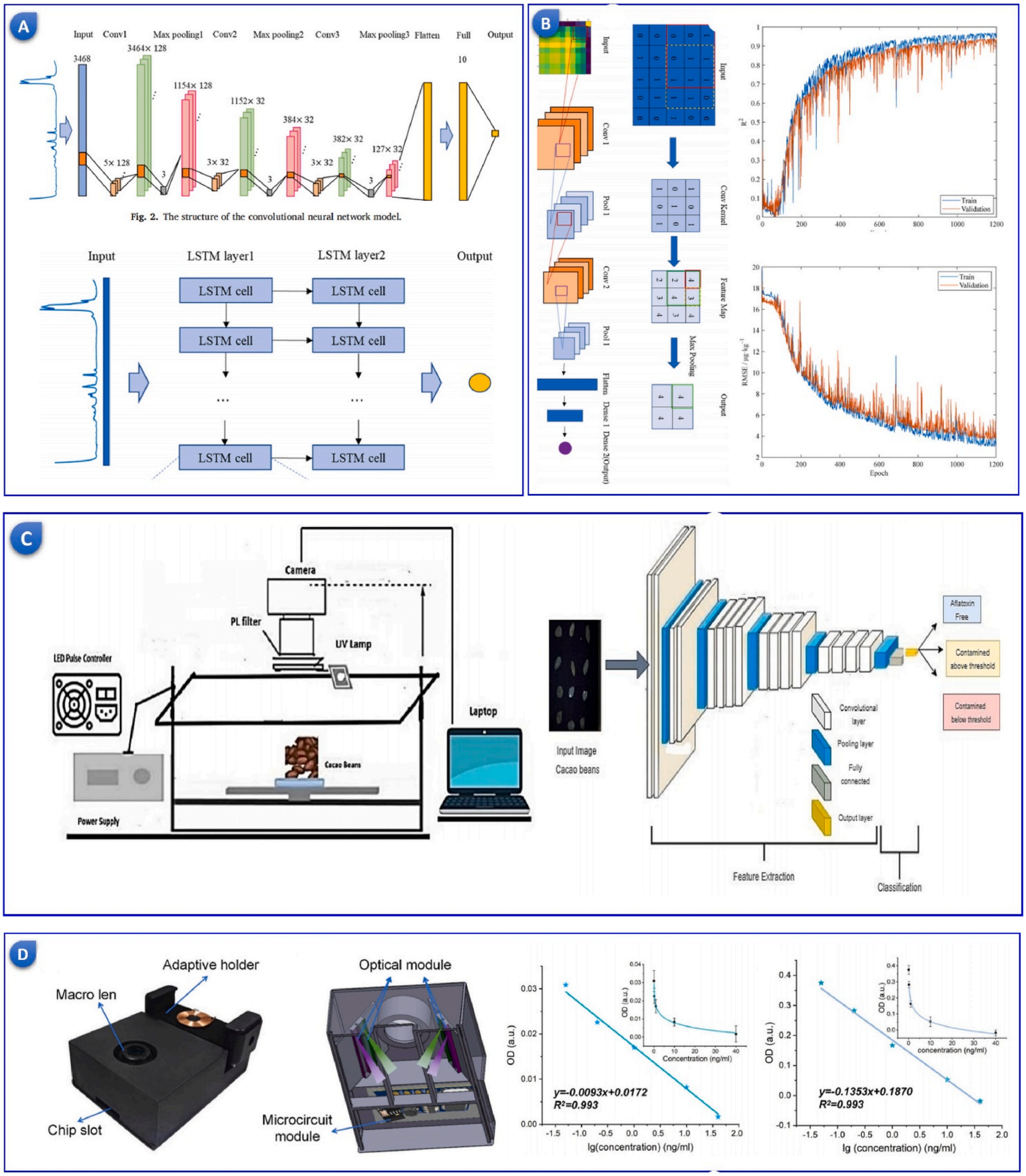

Fig. 5. A) Raman spectra-based deep learning models to detect aflatoxin B1 in edible oil, taken from (Deng, Zhang, et al., 2022) with the permission of Elsevier. B) Near-infrared spectra with two-dimensional CNN to detect aflatoxin, taken from (H. Jiang et al., 2023) with the permission of Elsevier. C) Aflatoxin detection in cocoa beans using fluorescence imaging and deep learning, taken from (Sadimantara et al., 2024), with the permission of Elsevier JRC. D) A smartphone-integrated deep learning aflatoxin detection, taken from (Sun et al., 2025) with the permission of Elsevier.

commodities is a serious impediment to the development of effective and efficient tools for use across multiple food crops for aflatoxin contamination detection (Castano-Duque et al., 2022). Researchers are investigating multi-modal learning strategies that can incorporate spectral, visual, or chemical data to improve model transferability. Such models will help facilitate the acquisition of shared aflatoxin signatures, regardless of the sample type. In addition to multi-modal strategies, meta-learning, and domain adaptation are likely to allow for more rapid adaptation to new datasets with few retraining iterations (Tang et al., 2023). Lastly, combining physics-informed ML models where the biological or chemical behavior is accommodated into the ML learning process should improve model generalization.

4.3. Regulatory transparency and the "Black Box" problem

Deep learning models may possess high accuracy and predictive power, yet deep learning models are often difficult to interpret. Due to this lack of interpretability, food regulators and safety authorities responsible for risk management and compliance may hesitate to trust or use it in decision-making. Establishing trust among stakeholders, regulators, and inspectors relies on transparency in methods and processes to inform judgements that impact trade, human health, and risk-based

Table 1
Comparative Table of different Aflatoxin case studies.

Case Study	Detection Technique	ML Model/Tool Used	Accuracy/R^2/LOD	Overarching Trend	Target Sample Type
Bertani et al. (2020)	Fluorescence spectroscopy	SVM	Accuracy: 94 %	Non-destructive portable detection	Almonds
Ataş et al. (2012)	Hyperspectral imaging	MLP, LDA with feature selection (PCA, SVM-RFE)	Up to 90 %	Compact vision system with feature selection	Chili Pepper
(X. Wang, Bouzembrak, Oude Lansink et al., 2022)	Risk-based monitoring with historical data	XGB, SVM, Decision Tree, Logistic Regression	Accuracy >90 %, AUC >0.8	Cost-effective ML-based surveillance	Feed Products
Han and Gao (2019)	Hyperspectral imaging	CNN (deep learning)	Pixel-level: >95 %	High-resolution pixel-based classification	Peanuts
(Y. K. Kim et al., 2023)	Raman spectroscopy	LDA, LSVM, QDA, PLSR	Accuracy: 95.7 %, R^2: 0.9998 (SNV)	Field-portable Raman systems	Ground Maize
Zhu et al. (2024)	Hyperspectral imaging	MSGhostDNN (Deep Learning)	Accuracy: 97.87 %	Contrastive learning with key wavelength mining	Peanuts, Corn
Deng et al. (2023)	Microwave sensing	CNN with multitask learning	Accuracy: 100 %, R^2: 0.9807	Multi-task detection and model fusion	Wheat
Zhu, Yang, and Han (2023)	Hyperspectral imaging	1D CNN, Dual-branch CNN	Accuracy: 91.3 %, R: 0.77	Subpixel key wavelength mining	Peanuts
Mei et al. (2023)	Accelerometry	GBDT, KNN, SVC, RF, DT	Accuracy: up to 97 %	Behavioral biometrics for toxin exposure	Broiler Chickens
Deng, Zhang, et al. (2022)	Raman spectroscopy	CNN, RNN	R^2: 0.95, RPD: 4.86	Quantitative Raman chemometrics	Edible Oil
Zhu et al. (2025b)	Hyperspectral imaging with deep spectral reconstruction	LSTM, PCA, Compressed Projection	Accuracy: 98.3 %, RMSE: 0.0214	Efficient detection with reduced model complexity	Peanut
Kılıç et al. (2024)	A UV-optical system with feature extraction	MobileNetV2, ResNet101V2, InceptionResNetV2 + SVM	Accuracy: 96 %	Transfer learning under UV imaging	Dried Figs
(W. Jiang et al., 2024)	Terahertz wave imaging	CNN	Accuracy: 98.7 %	Non-destructive seed quality check	Peanut
(C. Wang et al., 2025)	Hyperspectral imaging	MVO optimized CNN-BiLSTM	Accuracy: 94.92 %, Recall: 95.59 %	Model fusion with evolutionary optimization	Peanut
Zhu, Yang, Gao, et al. (2022)	Subpixel decomposition in hyperspectral imaging	ResNet18, LeNet5, AlexNet, VGG16	R^2: 0.8898, RMSE: 0.0138, RPD: 2.8851	Pixel-level regression for toxin quantification	Peanut
Kılıç and İnner (2022)	UV imaging	DenseNet169 (transfer learning)	Accuracy: 98.57 % (training), 97.5 % (validation)	Transfer learning in UV-based fig detection	Dried Figs
Sadimantara et al. (2024)	Fluorescence imaging	GoogLeNet, SqueezeNet, AlexNet, ResNet50	Accuracy: 96.42 %	CNN-based aflatoxin level classification	Cocoa Beans
Castano-duque et al. (2025)	Remote sensing, satellite, and weather data	Gradient Boosting, Neural Network (Ratkowsky-ARI)	Accuracy: 73 %, Sensitivity: 71 %	Geospatial ML for outbreak forecasting	Corn (Texas)
Guo et al. (2025)	Hyperspectral imaging	Deep Modular Combination Optimization (DMCO)	R^2: 0.879, RMSE: 1.269	Hybrid modular DL model optimization	Peanut Kernels
(H. Jiang et al., 2023)	Near-infrared spectroscopy with GASF coding	2D CNN (GASF-2D-CNN)	R^2: 0.99, RMSEP: 2.0 μg/kg, RPD: 8.3	Spectral image encoding improves regression performance	Peanut
Zhongzhi and Limiao (2020)	Hyperspectral imaging with UV-induced fluorescence	RBF-SVM with Fisher optimization	R: 0.9785, Accuracy: 95.5 %	Radiation index features with narrowband SVM	Peanut
Akdemir Evrendilek et al. (2022)	Pulsed Electric Field treatment data	GBRT, RFR, ANN	R^2 (GBRT inactivation): high; RMSE: lowest	Energy-level regression for toxin inhibition	Red Pepper Flakes
(B. Wang et al., 2022)	Near-Infrared Spectroscopy	1D-CNN, 2D-MTF-CNN	R^2: 0.9955, RMSE: 1.3591	Encoding NIR as a 2D image improves CNN performance	Maize
Zhu, Zhao, et al. (2023)	Hyperspectral imaging	Fuzzy C-means, Random Forest + pre-trained CNN	Accuracy: 97.11 %, Clustering Accuracy: 95.51 %	Unsupervised + transfer learning fusion	Peanut
Sabo et al. (2024)	GC-MS of plant-based bVOCs	Random Forest, LDA	F1 Score (RF): 0.80	Volatile-based biomarker detection	Peanut plants, pods, kernels
Manhando et al. (2021)	Optical Coherence Tomography	DCNN feature extractor + ECOC-SVM	Accuracy: 85 % (48h), 96 % (96h)	Early mold detection from OCT images	Peanut
Branstad-Spates et al. (2023)	Geospatial prediction using weather/satellite	Gradient Boosting Machine (GBM)	Accuracy: 90.32–96.77 %, Sensitivity: Low	Forecasting aflatoxin via remote sensing	Corn (Iowa)
Aref et al. (2024)	Data modeling from demographic/environmental factors	Random Forest, Gradient Boosting	Qualitative outcome not fully quantified	Health-focused ML for infant food safety	Breast Milk
Zhu, Yang, Ding, and Han (2022)	Hyperspectral imaging	1D-Modified Temporal Convolutional Network	Train: 99.60 %, Test: 99.26 %	Pixel-level hyperspectral deep regression	Peanut
Li et al. (2023)	Near-Infrared Spectroscopy	SVM with BWO-IVSO feature selection	R^2: 0.9761, RMSEP: 24.63 μg/kg, RPD: 4.70	Portable NIR + iterative variable optimization	Peanut
Gao et al. (2021)	Hyperspectral Imaging	1D CNN	Accuracy: 96.35 % (Peanut), 92.11 % (Maize)	Fast CNN-based pixel classification	Peanut, Maize
Madushan et al. (2024)	Multispectral Imaging	LDA, Ensemble Classifiers	Accuracy: 90 %, R^2: 0.9723	Cost-effective multispectral food monitoring	Bread

(*continued on next page*)

Table 1 (*continued*)

Case Study	Detection Technique	ML Model/Tool Used	Accuracy/R^2/LOD	Overarching Trend	Target Sample Type
(Y. Kim et al., 2024)	CO_2 Respiration + Visual Inspection	Compact Convolutional Transformer (CCT)	Accuracy: 83.33 %, F1-score: 0.95 (contaminated)	AI for early mold detection via respiration	Wheat
Hu, Zhan, Chen, et al. (2023)	THz-TDS with metamaterial sensor	CARS-RBF-SVM	Accuracy: 99.17 %	Trace-level qualitative toxin ID	Aflatoxin solutions (B2, G1, G2)
Bertani et al. (2024)	Fluorescence Imaging + RGB to Y'CbCr	NN with ROC-based feature reduction	Accuracy: 84.7–93 %	Color space + ML for portable detection	Almonds
Deng, Jiang, and Chen (2022)	Raman Spectroscopy	SVM with CARS	R^2: 0.9715, RMSEP: 3.54 μg/kg, RPD: 5.83	Portable Raman + feature optimization	Maize
Sun et al. (2025)	Dual-modal Immunochromatographic Chips	YOLOv5	Accuracy: 98 %, LOD: 2.2 pg/mL	Smartphone-integrated toxin detection	Aflatoxin B1 (general food matrices)
Kabir et al. (2024)	Hyperspectral Imaging	Multilayer Perceptron (MLP)	R^2: 0.95+, high accuracy	Hybrid spectral band selection	Peanut
Zhu et al. (2025a)	Hyperspectral Imaging + LLM	Fine-tuned Babbage-002 LLM	R^2: 0.8898, RMSE: 0.0588, RPD: 2.74	LLM-driven spectral regression	Peanut
(S. Y. Zheng et al., 2020)	Near-Infrared Reflectance Spectroscopy	XGBoost + SVM	RMSEP: 3.57 μg/kg, Accuracy: 90.32 %	Early AFB1 prediction via Ver A monitoring	Maize
Ranbir et al. (2025)	Metal oxide-based sensor array	LDA (multivariate analysis)	Accuracy: 100 %, LOD: 0.02–0.09 ppm	On-site multi-toxin classification	Corn
Hu, Zhan, Wang, et al. (2023)	THz metamaterial sensor	Grouped regression model	LOD: 7.28×10^{-11} to 1.22×10^{-7} mg/mL	Sub-model grouping for high sensitivity	Aflatoxin B2 Solution
Williams et al. (2025)	Hyperspectral Imaging	ResNet, VAE, DCGAN, K-Means	ResNet: 96.67 %, K-Means: 84.38 %	Single wavelength ML classification	Pistachio Nuts
Castano-Duque et al. (2022)	Satellite + Meteorological data	Gradient Boosting, Bayesian Network	Accuracy: 94 %	Seasonal prediction of mycotoxin risk	Maize (Illinois)
Smeesters et al. (2023)	Fluorescence Spectroscopy	Spectral Separation + Classification	Sensitivity: μg/kg level (0.6–1647.8)	Portable device for in-field detection	Maize
Camardo Leggieri et al. (2021)	Meteorological + Cropping data	Deep Neural Network (DNN)	Accuracy >75 %	Weather + crop system prediction model	Maize (Italy)

compliance. Recent advancements in Explainable AI (XAI) are addressing this transparency gap by supplying visual and textual explanations for model-produced predictions. For instance, saliency maps in image-based detection models depict the respective regions of an image that informed and ultimately resulted in a decision of contamination (Thalpage, 2023). Similarly, attention mechanisms in spectral models identify key wavelength intervals contributing to a decision. These approaches can cultivate trust from stakeholders and regulators to agree upon approval; they can also be used to educate non-expert users.

4.4. Real-time monitoring in supply chains

Conventionally designed AI models are suitable for controlled conditions in lab-based contexts. On the contrary, genuine supply chains from farm to warehouse vastly change with dynamic conditions outside of environmental controls, power constraints, and limited outside connections (Helo & Shamsuzzoha, 2020). The combination of IoT with AI provides real-time, continuous environmental and contamination monitoring parameters for real supply chains. Utilization of lightweight edge-AI models can significantly decrease latency and the use of cloud technology by processing sensor data locally (Abdelhamid et al., 2024). Using reinforcement learning, IoT/AI systems can change storage conditions (e.g., humidity, airflow) to limit fungal growth and toxin accumulation, transforming passive storage conditions into intelligent, controllable environments.

4.5. Lack of preventive solutions beyond detection

Most AI-generated developments for aflatoxin management target the post-harvest detection of aflatoxins. However, contamination has already occurred, and economic or public health harm may be irreversible. In the future, AI and CRISPR-based gene editing will be used to develop aflatoxin-resistant crop varieties. AI will sift through massive omics datasets to discover genes corresponding to either mold resistance or suppression of aflatoxin biosynthesis (NavyaH et al., 2020). When the targets of concern are determined, CRISPR can edit the genomes of plants at the target site to insert qualities or traits of interest. This preventative tactic could substantially lessen aflatoxin before harvest, especially for tropical crops grown in aflatoxin-risk areas.

4.6. Limited consumer-level safety solutions

Despite improvements in laboratory and industrial testing methods, consumers cannot assess aflatoxin contamination in food credibly, efficiently, and affordably at the point of use. The emergence of AI-enhanced smart-packing and APP-based detection technologies represents a paradigm shift in consumer participation and empowerment in food safety efforts. Enhanced food packaging materials with nanosensors embedded in the packaging matrix may detect chemical changes in the packaging matrix and relay information and alerts to an APP on consumers' smartphones (Mavani et al., 2022). AI models process electrical signals from the nanosensors and produce a user-friendly display (e.g., color change of a sensing zone, report via QR code) and guide user action (i.e., use immediately, cook or dispose of food). In this way, these smart packaging innovations democratize aflatoxin monitoring and improve food safety during last-mile delivery and household storage. By tackling the overlapping social, technical, and policy challenges with scalable AI solutions, collaborative data-sharing efforts, and cross-disciplinary innovation, the future of aflatoxin detection will be moved beyond reactive to predictive and preventive detection. As questions of explainability, portability, and integration are relatively resolved in AI technologies, AI can potentially disrupt food safety paradigms for healthier lives, resilient agricultural economies, and food system sustainability globally (Y. Zheng et al., 2021).

5. Conclusion

AI has risen as a disruptor in detecting and managing aflatoxins, offering unmatched accuracy, speed, and scalability. By synergizing AI with spectral, imaging, and behavioral data, AI-enabled systems have outperformed traditional detection and management methods in laboratory and field settings. This review indicated the varied AI model types, from supervised classifiers and regressors to deep learning architectures and reinforcement learning agents, coupled with applicable case studies demonstrating their effectiveness in real food systems. While challenges remain, including data shortages, model generalization, and regulatory traceback, advances in explainable AI, edge computing, and innovative packaging herald a transition toward mass adoption of accessible, predictive, and preventive solutions. As the agri-food industry embraces these innovations, AI sits on the cusp of transforming and modernizing global food safety, reducing potential health-based harms, and enabling resilient, sustainable food systems.

Availability of data and material

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Declaration of competing interest

The authors declare no conflict of interest.

Data availability

Data will be made available on request.

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